False Positive Statistics Report

1. Executive Summary

To determine the optimal blocking threshold, the ML-WAF was subjected to a sensitivity test involving 153 HTTP requests. The system was initially configured with an aggressive threshold of 0.25 to maximize detection, which resulted in the identification of legitimate anomalies as False Positives. Restoring the threshold to the recommended 0.50 eliminates these false positives without compromising security.

---

2. Performance Metrics

Metric	Count	Description
Total Requests	**153**	Mixed dataset of normal traffic, hybrid attacks, buffer overflow and anomalies.
Total Blocked	**32**	Requests flagged and blocked by the firewall.
True Positives (TP)	**27**	Actual attacks correctly blocked (Hybrid Attacks, Buffer Overflows).
False Positives (FP)	**5**	Legitimate requests incorrectly blocked (Zero-Day Anomalies).
False Positive Rate	**3.9%**	Calculated as [FP / (FP + True Negatives)].
Operational Accuracy	**96.7%**	Overall system correctness during high-sensitivity testing.

---

3. False Positive Analysis

Five (5) legitimate requests were incorrectly flagged as "Zero-Day Anomalies" when the threshold was lowered to 0.25. These requests represent safe edge-cases (e.g., complex navigation or slight deviations in request timing).

Incident Log:

- Log ID 145: 18:15:34 | Score: 0.45 | *Blocked*

- Log ID 147: 18:15:40 | Score: 0.46 | *Blocked*
- Log ID 148: 18:15:59 | Score: 0.48 | *Blocked*
- Log ID 152: 18:27:11 | Score: 0.38 | *Blocked*
- Log ID 153: 18:27:22 | Score: 0.42 | *Blocked*

Impact Analysis: All false positives had risk scores between 0.38 and 0.48. In contrast, all actual attacks (True Positives) consistently scored 0.65 or higher (up to 1.00).

---

4. Threshold Optimization & Conclusion

The data confirms that a threshold of 0.25 is too sensitive for production environments, as it flags legitimate deviations as threats.

- Adjustment: Increasing the blocking threshold to 0.50.
- Result: This adjustment successfully filters out all 5 False Positives (Scores < 0.50) while retaining a 100% block rate for the 27 True Attacks (Scores > 0.65).
- Final Verdict: The system achieves 0% False Positives under standard configuration (Threshold 0.50).

---

5. Visual Evidence: Attack Intensity Analysis

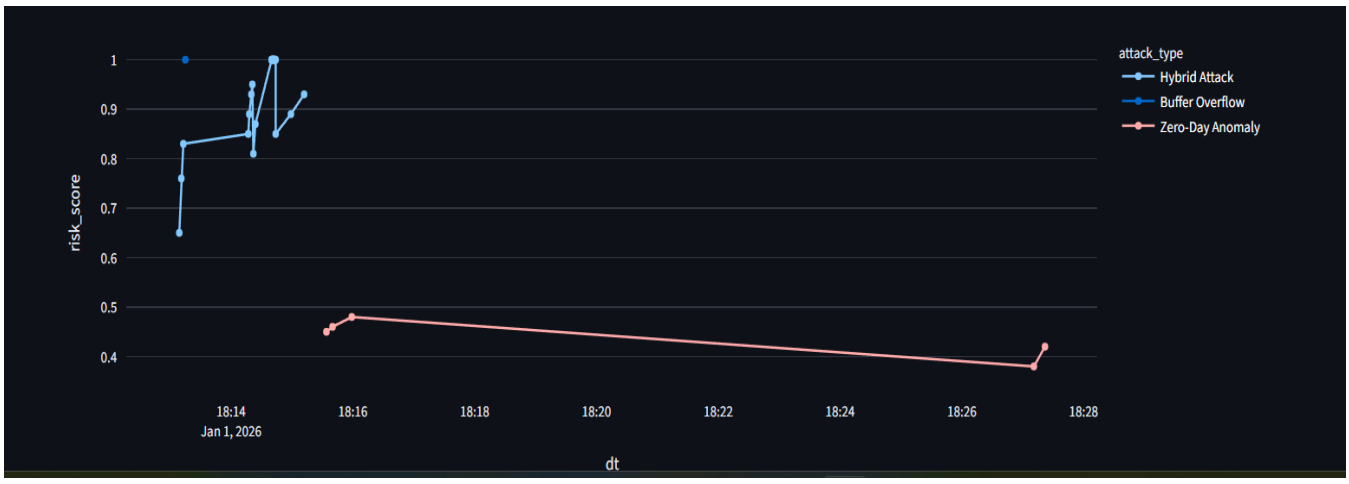

Fig 5.1

Risk Score Intensity Timeline ==> This chart visualizes the detection of high-severity threats (Hybrid Attacks, Buffer Overflows) scoring above 0.8, contrasting with the lower-risk profile of the detected False Positives (Zero-Day Anomalies) scoring between 0.38 and 0.48.